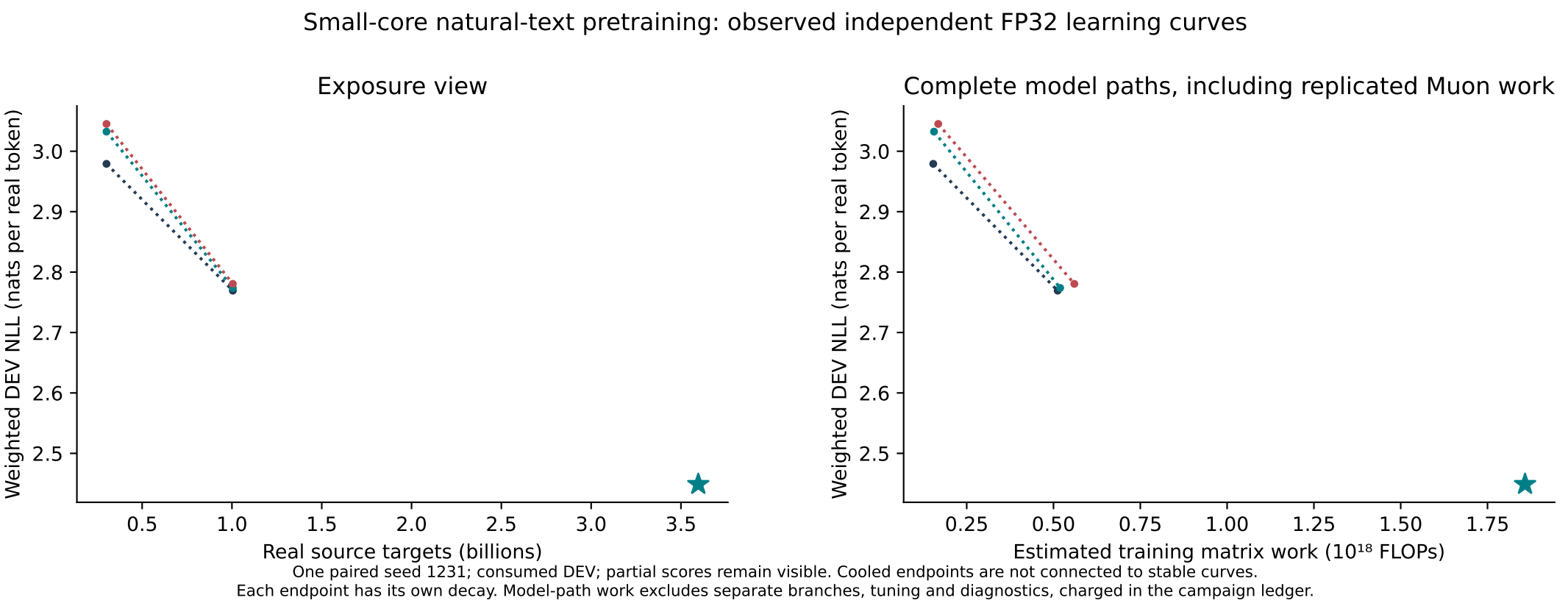
Small-core natural-text pretraining: observed independent FP32 learning curves
Exposure view
Complete model paths, including replicated Muon work
Weighted DEV NLL (nats per real token)
3.0
2.9
2.8
2.7
2.6
2.5
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Real source targets (billions)
0.25
0.50
0.75
1.00
1.25
1.50
1.75
Estimated training matrix work (10¹⁸ FLOPs)
One paired seed 1231; consumed DEV; partial scores remain visible. Cooled endpoints are not connected to stable curves.
Each endpoint has its own decay. Model-path work excludes separate branches, tuning and diagnostics, charged in the campaign ledger.
Ordinary
True one-pass
Sampled recurrence, exit 2
Stable LR
Cooled equal-data endpoint
Cooled matched-work endpoint